

How to handle space groups

9

9.1 Wyckoff positions of space groups

The (infinite) set of symmetry-equivalent points in a space group $\mathcal{G}$ is called a $\mathcal{G}$ -orbit or *crystallographic point orbit* (also point configuration), cf. Definitions 5.11 and 5.12 (page 60) [99–102]. If it is clear that the subject in question concerns points of a space group, for short, we simply call it an orbit or point orbit. If the coordinates of a site are completely fixed by symmetry (e.g. $\frac{1}{4}, \frac{1}{4}, \frac{1}{4}$), then the orbit is identical with the corresponding *Wyckoff position* of the space group. However, if there are one or more freely variable coordinates (e.g. x in $x, \frac{1}{4}, 0$), the Wyckoff position comprises an infinity of possible orbits; they differ in the values of the variable coordinate(s), cf. Definition 6.6 (page 65) and Section 6.5 (page 81). For example, in space group $Pbcm$ the set of points that are symmetry equivalent to, say, $0.391, \frac{1}{4}, 0$, makes up one orbit. The set corresponding to $0.468, \frac{1}{4}, 0$ belongs to the same Wyckoff position $4c$ of $Pbcm$, but to another orbit (its variable coordinate x is different).

It is customary to designate a Wyckoff position of a space group by its *Wyckoff symbol*. It consists of the multiplicity and the Wyckoff letter, for example, $4c$, see Section 6.4.2, page 79.

A consequence of this kind of designation is the dependence of the multiplicity on the size of the chosen unit cell. For example, the multiplicities of rhombohedral space groups are larger by a factor of 3 if the unit cell is not referred to rhombohedral, but to hexagonal axes.

Many space groups have different Wyckoff positions with the same type of point symmetry; combined they form a Wyckoff set (called *Konfigurationslage* by [99]). These Wyckoff positions are mapped onto one another by the *affine* normalizer of the space group (cf. Section 8.2, page 105).

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Example 9.1

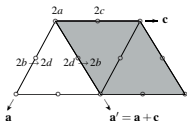
Space group No. 23, $I222$, has six Wyckoff positions with the site symmetry 2; together they make up one Wyckoff set:

$4e(x, 0, 0)$ and $4f(x, 0, \frac{1}{2})$ on twofold rotation axes parallel to **a**,
 $4g(0, y, 0)$ and $4h(\frac{1}{2}, y, 0)$ on twofold rotation axes parallel to **b**,
 $4i(0, 0, z)$ and $4j(0, \frac{1}{2}, z)$ on twofold rotation axes parallel to **c**.

However, if the lattice parameters are $a \neq b \neq c$, in the preceding example the positions $4e, 4f$ cannot be considered to be equivalent to the positions $4g, 4h$ and to $4i, 4j$, being on differently oriented axes. On the other hand, the positions $4e$ and $4f$ are equivalent; they are mapped onto each other by the *Euclidean* normalizer (they are equivalent in the Euclidean normalizer).

As explained in Section 8.5, as a rule, there exist several equivalent sets of coordinates that describe the very same crystal structure; they can be interconverted with the aid of the Euclidean normalizer. The change from one to another coordinate set can imply an interchange of Wyckoff positions among the positions that are equivalent in the Euclidean normalizer. If the origin of space group $I222$ is shifted by $0, 0, \frac{1}{2}$, the atoms at the Wyckoff position $4e$ are shifted to $4f$ and vice versa. This is similar to the interchange of Na and Cl atoms in Example 8.3 (page 111).

With monoclinic space groups, different unit cells can be chosen for the same structure; their bases can be interconverted by transformations such as $\mathbf{a} \pm n\mathbf{c}$, $\mathbf{b}$, $\mathbf{c}$ (n = integral number). This can also cause an interchange of Wyckoff symbols. The same applies to basis transformations of the space group $P\bar{1}$.



Example 9.2

The space group $P12_1/c1$ has four kinds of inversion centres:

$$2a, 0, 0, 0; \quad 2b, \frac{1}{2}, 0, 0; \quad 2c, 0, 0, \frac{1}{2}; \quad 2d, \frac{1}{2}, 0, \frac{1}{2}$$

By transformation to a cell with the basis $\mathbf{a} + \mathbf{c}$, $\mathbf{b}$, $\mathbf{c}$ (grey in the image in the margin), the Wyckoff positions $2a$ and $2c$ keep their Wyckoff symbols, whereas the other two are interchanged $2b \rightleftharpoons 2d$.

9.2 Relations between the Wyckoff positions in group–subgroup relations

For every group–subgroup relation $\mathcal{G} \rightarrow \mathcal{H}$ it is essential to keep track of which Wyckoff positions of the subgroup $\mathcal{H}$ result from those Wyckoff positions of the space group $\mathcal{G}$ that are occupied by atoms. A group–subgroup relation can only be correct and fraught with meaning if there exist clear connections for all atoms.

The atomic positions show clearly how the symmetry is being reduced step by step in a sequence of group–subgroup relations. At first, as a rule, the atoms occupy special positions, i.e. they are located on certain symmetry elements with fixed values for the coordinates and with some specific site symmetry. When proceeding from group to subgroup, the atomic positions experience the following changes one by one or jointly [103]:

- (1) some or all coordinates x, y, z fixed or coupled by symmetry become independent, i.e. the atoms can shift away from a special position;
- (2) the site symmetry is reduced;
- (3) the point orbit splits into different orbits.

If the index of the symmetry reduction is 2, either the site symmetry is kept and the orbit splits or there is no splitting and the site symmetry is reduced.

There is a one-to-one relation between the points of an orbit and the corresponding points of a subgroup. They comprise the same number of points in the same volume. The multiplicity of a Wyckoff position shows up in the multiplicities of the corresponding Wyckoff positions of the subgroup. If the

unit cell selected to describe the subgroup has the same size, the sum of the multiplicities of the positions of the subgroup must be equal to the multiplicity of the position of the starting group. For example, from a position with multiplicity of 6, a position with multiplicity of 6 can result, or it can split into two positions of multiplicity of 3, or into two with multiplicities of 2 and 4, or into three with multiplicity of 2. If the unit cell of the subgroup is enlarged or reduced by a factor f , then the sum of the multiplicities must also be multiplied or divided by this factor f .

Between the positions of a space group and those of its subgroups there exist unique relations for given relative positions of their unit cells. Usually, there are several possible relative positions; the relations between the Wyckoff positions may differ for different (arbitrary choices of) relative positions of the origins of the group and the subgroup.

The relations between the Wyckoff positions of the space groups and their subgroups can be derived from the data of *International Tables A*. However, this is a cumbersome task prone to errors. Therefore, it is recommended to use the tables of *International Tables A1*. Listed are all maximal subgroups of all space groups and what Wyckoff positions of a subgroup result from the Wyckoff positions of each space group. The listed relations in each case *are valid only for the given basis transformation and origin shift*. With other basis transformations or origin shifts the Wyckoff symbols of the subgroup may have to be interchanged among the positions that are equivalent in the Euclidean normalizer. In addition, the computer program WYCKSPLIT can be used; it computes the relations after input of the space group, subgroup, basis transformation, and origin shift [104].

9.3 Non-conventional settings of space groups

As a rule, it is recommended to describe crystal structures with the conventional settings of the space groups and taking into account the standardization rules mentioned in Section 8.4. However, standardized settings may mean that related crystal structures have to be described differently, with the consequence that the similarities become less clear and may even be obscured. For the comparison of crystal structures, it is preferable to set up all unit cells in a most uniform way and to avoid cell transformations as far as possible, even if this requires the utilization of non-conventional settings of the space groups. In this section some instructions are given referring to such non-conventional settings.

9.3.1 Orthorhombic space groups

Among orthorhombic space groups it is frequently suitable to choose settings that deviate from the listings in *International Tables A*. This is due to the fact that there exists no special preference for any of the axes directions **a**, **b**, and **c** in the orthorhombic system. Generally, there are six possible settings. All six are mentioned in Table 4.3.2.1 of *International Tables A*, but only one of them is considered as the standard setting and has been completely listed. The other five are obtained by interchange of the axes, namely:

- (1) Conventional setting: **abc**
- (2) Cyclic interchange 'forwards': **cab**
- (3) Cyclic interchange 'backwards': **bca**
- (4) Interchange of **a** and **b**: **ba $\bar{c}$** or **b $\bar{a}$ c** or **$\bar{b}$ ac**
- (5) Interchange of **a** and **c**: **cb $\bar{a}$** or **c $\bar{b}$ a** or **$\bar{c}$ ba**
- (6) Interchange of **b** and **c**: **ac $\bar{b}$** or **a $\bar{c}$ b** or **$\bar{a}$ cb**

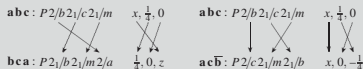
The notation **cab** means: the original **a** axis is now in the position **b**, etc., or: convert **a** to **b**, **b** to **c**, **c** to **a**. The options 4 to 6 (interchange of two axes) require the reversal of the direction of one axis in order to retain a right-handed coordinate system.

The interchange of the axes has the following consequences:

- (1) The lattice parameters *a*, *b*, *c* must be interchanged.
- (2) In the Hermann–Mauguin symbol the sequence of the symmetry directions has to be changed.
- (3) The notations of the glide directions *a*, *b*, *c* in the Hermann–Mauguin symbol have to be interchanged. *m*, *n*, *d*, and *e* remain unaltered.
- (4) The notations of the centrigs *A*, *B*, *C* have to be interchanged. *P*, *F*, and *I* remain unaltered.
- (5) In the coordinate triplets of the atomic positions the sequence and the notations have to be interchanged. If the direction of an axis is reversed, the signs of the corresponding coordinates have to be inverted. Usually, the Wyckoff symbols remain unaltered. However, caution is required if the interchange of axes concerns space groups whose symbols do not reveal that or how the axes have been interchanged (see Examples 9.5 and 9.6).
- (6) The notations of rotations and screw rotations do not change.

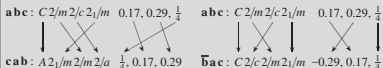
Example 9.3

Two possibilities to interchange axes of the space group $P2/b2_1/c2_1/m$ (*Pbcm*, No. 57) and its Wyckoff position $4c(x, \frac{1}{4}, 0)$:



Example 9.4

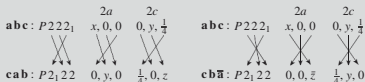
Two possibilities to interchange axes of the space group $C2/m2/c2_1/m$ (*Cmcm*, No. 63) and its Wyckoff position $8g(x, y, \frac{1}{4})$ with $x = 0.17$ and $y = 0.29$:



In most cases the non-conventional Hermann–Mauguin symbol shows in a unique way how the setting differs from the conventional setting. However, there are exceptions, as shown in the following two examples.

Example 9.5

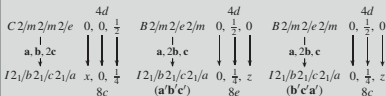
In the space group $P222_1$ (No. 17) the Wyckoff position $2a$ ($x, 0, 0$) is situated on twofold rotation axes parallel to **a**, and the position $2c$ ($0, y, \frac{1}{4}$) on twofold rotation axes parallel to **b**. The following two possibilities to interchange axes result in the same non-conventional space-group symbol:



After cyclic interchange the twofold rotation axes of the Wyckoff position $2a$ are parallel to **b** at $0, y, 0$. On the other hand, interchange of **a** and **c** yields twofold rotation axes parallel to **b** at $\frac{1}{4}, y, 0$. The non-conventional symbol $P2_122$ does not show how the basis vectors have been interchanged and where the rotation axes are situated. In such cases cyclic interchange should be preferred.

Example 9.6

The space group $Cmme$ ($C2/m2/m2/e$, No. 67) has a subgroup $Ibca$ ($I2_1/b2_1/c2_1/a$, No. 73) with doubled **c** vector. The Wyckoff position $4d$ ($0, 0, \frac{1}{2}$) of $Cmme$ becomes $8c$ ($x, 0, \frac{1}{4}$) of $Ibca$ with $x \approx 0$. Starting from $Bmem$, a non-conventional setting (**bca**) of $Cmme$, one obtains the same subgroup $Ibca$ by doubling **b**. The position $4d$ of $Bmem$ is ($0, \frac{1}{2}, 0$), from which one obtains ($0, \frac{1}{4}, z$) of $Ibca$ ($z \approx 0$). According to *International Tables A*, ($0, \frac{1}{4}, z$) is not the position $8c$, but $8e$ of $Ibca$; therefore, it has another denomination than when starting from $Cmme$. However, one can choose the non-conventional setting **bca** for $Ibca$, which results in the Wyckoff symbol $8c$ for the position ($0, \frac{1}{4}, z$). The symbol $Ibca$ does not reveal this because cyclic interchange of the axes does not change the symbol. In this case the interchanged axes and Wyckoff symbols must explicitly be mentioned to avoid confusion.



9.3.2 Monoclinic space groups

The interchange of axes of monoclinic space groups has the same consequences as for orthorhombic space groups. In addition, the interchange of the angles α, β, γ has to be considered. *International Tables A* list the settings with **b** and

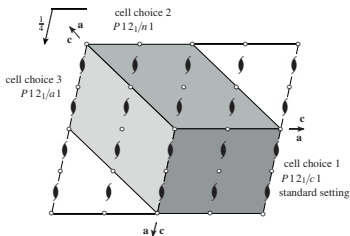


Fig. 9.1 The three cell choices for space group No. 14, $P2_1/c$, with monoclinic b axis.

c as monoclinic axis. Settings with monoclinic a axis can be obtained from these settings in the same way as described in the preceding section.

A few additional particularities have to be taken into account in the case of monoclinic space groups with centring or with glide planes. Three cell choices have each been listed for these space groups (cell choices 1, 2, and 3). The direction of the monoclinic axis and the cell choice can be uniquely recognized from the full Hermann-Mauguin symbol (Fig. 9.1). If a setting other than the standard setting is chosen ($P2_1/c$ 1 in Fig. 9.1), the full symbol must be stated. A change of the cell choice not only entails a change of the Hermann-Mauguin symbol, but also of the monoclinic angle.

The interchange of axes of the space groups Cc and $C2/c$ requires special care. An interchange of the axes a and c (keeping the cell and $-b$ as monoclinic axis), results in the following change of the space group symbol:

$$C12/c\ 1 \text{ (cell choice 1)} \longrightarrow A12/a\ 1$$

$A12/a\ 1$ is not a conventional setting. One of the settings listed in *International Tables* is $A12/n\ 1$ (cell choice 2). $A12/n\ 1$ and $A12/a\ 1$ refer to the same space group with the same cell, but with shifted origin positions. The space group has a as well as n glide planes; the one mentioned in the symbol is located at $y = 0$, the other one at $y = \frac{1}{4}$. The change of the setting from $A12/a\ 1$ to $A12/n\ 1$, keeping the axes directions, involves an origin shift by $0, \frac{1}{4}, \frac{1}{4}$ (Fig. 9.2).

Cyclic interchange of the axes $abc \rightarrow cab$ yields a change of the monoclinic axis from b to c . $C12/c\ 1$ is then converted to $A112/a$. One can also interchange b and c : $abc \rightarrow acb$; in this case the position of the origin must be watched. $C12/c\ 1$ is then converted to $B112/b$ with glide planes b at $z = 0$ and n at $z = \frac{1}{4}$. $B112/b$ had been the setting used for the monoclinic c axis in the old editions of *International Tables* (up to 1969). However, since the 1983 edition it has been $B112/n$ (monoclinic c , cell choice 2), with glide planes n at $z = 0$ and b at $z = \frac{1}{4}$. $B112/b$ and $B112/n$ differ in their origin positions by $\frac{1}{4}, 0, \frac{1}{4}$. The denominations of the Wyckoff symbols in the old editions of *International Tables* (up to 1969) differ from the newer editions.

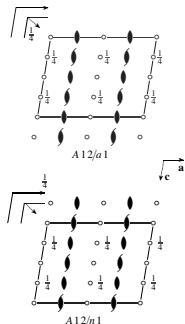
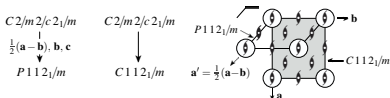


Fig. 9.2 The two settings $A12/a\ 1$ and $A12/n\ 1$ of the same space group with the same axes directions differ in the positions of their origins.

If one can choose, non-conventional monoclinic settings that result from cyclic interchange of axes should be preferred. In that case the coordinate triplets have to be interchanged correspondingly, but there are no origin shifts. The Wyckoff letters remain unchanged. In addition, the monoclinic angle keeps its value (β for monoclinic **b** axis). If **a** and **c** are interchanged, β only keeps its value if the direction of **b** is reversed, whereas the settings **cba** and **ba** require a replacement of β by $180^\circ - \beta$.

Sometimes it is useful to choose a non-conventional centred setting. For example, the space group $Cmcm$ has a subgroup $P112_1/m$ that can also be set as $C112_1/m$. This setting avoids a cell transformation:



A disadvantage of non-conventional centring is the lack of tables of the coordinate triplets for the Wyckoff positions. Wyckoff symbols are only unique if the corresponding coordinate triplets are explicitly stated.

9.3.3 Tetragonal space groups

Sometimes it is useful to choose *C*-centred instead of primitive settings or face-centred instead of body-centred settings of tetragonal space groups. Their a' and b' axes run diagonal to the axes of the conventional setting (Fig. 9.3). For example, this should be considered for a relation from a face-centred cubic space group to a tetragonal subgroup, which can then be described with unchanged axes and an *F* instead of a (conventional) *I* cell. This is advisable if a further subgroup follows which again has the metric of the original *F* cell; in this way two cell transformations can be avoided.

In the Hermann-Mauguin symbol of the non-conventional centred setting the symmetry elements referring to the directions **a** and **a - b** are interchanged. In addition to the letters for the centring, the following letters for glide planes have to be changed:

	setting		glide		glide	
	(centring)		planes $\perp c$		planes $\perp a$	
conventional	<i>P</i>	<i>I</i>	<i>a</i>	<i>n</i>	<i>b</i>	<i>n</i>
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
non-conventional	<i>C</i>	<i>F</i>	<i>d</i>	<i>e</i>	<i>g</i> ₁	<i>g</i> ₂

The glide planes g_1 and g_2 run perpendicular to $a' - b'$ and $a' + b'$ (in the non-conventional setting) and have glide components of $\frac{1}{4}, \frac{1}{4}, 0$ and $\frac{1}{4}, \frac{1}{4}, \frac{1}{2}$, respectively.

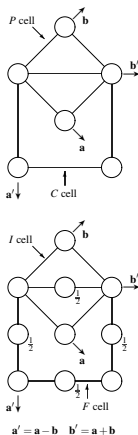


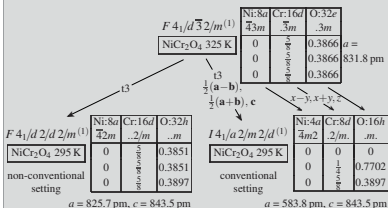
Fig. 9.3 Relative orientations of *C* and *F* centred tetragonal unit cells.

Example 9.7

Transformation of two tetragonal space groups to unit cells with doubled volume and diagonal **a** and **b** axes:

 $I4cm$  $F4mc$ $P4/n2/b2/m$  $C4/e2/m2/g_1$ **Example 9.8**

Above 47 °C, NiCr_2O_4 has the cubic spinel structure. Upon cooling, the structure experiences a slight distortion and becomes tetragonal [105]. If the tetragonal structure is not described with a body-centred but with a face-centred setting, the small amount of the distortion concerning the coordinates of the O atoms becomes much more apparent (tree of group–subgroup relations as explained in Chapter 10):



Transformation $I4_1/a2/m2/d \rightarrow F4_1/d2/d2/m$ according to $\mathbf{a} + \mathbf{b}$, $-\mathbf{a} + \mathbf{b}$, $\mathbf{c}$ and $\frac{1}{2}(x+y)$, $\frac{1}{2}(-x+y)$, z .

Note the sequence in the Hermann–Mauguin symbols, also at the site symmetry of the Ni atom, and the halved multiplicities at the Wyckoff symbols of the conventional setting $I4_1/a2/m2/d$ (because of the halved size of the unit cell).

A disadvantage as for all non-conventional centring is the lack of tables of the coordinates of the Wyckoff positions. One has to be specially careful when space groups with d glide planes are involved. For instance, for the space group $I4_1/amd$ ($I4_1/a2/m2/d$) mentioned in Example 9.8, it makes a difference whether the transformation $I4_1/amd \rightarrow F4_1/ddm$ is performed according to $\mathbf{a} + \mathbf{b}$, $-\mathbf{a} + \mathbf{b}$, $\mathbf{c}$ or $\mathbf{a} - \mathbf{b}$, $\mathbf{a} + \mathbf{b}$, $\mathbf{c}$, because the d glide directions are different. The latter transformation results in wrong glide directions unless an origin shift of $-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}$ is included in the group–subgroup relation.

The transformation of space groups that allow two origin choices can involve different origin shifts depending on the origin choice.

9.3.4 Rhombohedral space groups

Basis transformations concerning rhombohedral space groups can be rather complex. In addition, it has to be watched whether the setting refers to 'rhombohedral' or to 'hexagonal' axes. The transformations are notoriously prone to errors.

Every cubic space group has rhombohedral subgroups that have their three-fold axes parallel to one of the four directions of the body diagonals of the cube. Only in the case of primitive cubic space groups does the conventional cell of a maximal rhombohedral subgroup have the same unit cell, provided that rhombohedral axes have been chosen. The maximal rhombohedral subgroups of F - and I -centred cubic space groups can be derived without cell transformations if non-conventional F - or I -centred settings are chosen with rhombohedral axes (i.e. with $a = b = c$, $90^\circ \neq \alpha = \beta = \gamma \approx 90^\circ$). The space-group symbol then begins with $F\bar{3}$, $F3$, $I\bar{3}$, or $I3$, for example, $F\bar{3}m$ instead of $R\bar{3}m^{(rh)}$ (not to be confounded with $Fm\bar{3}$).

'Hexagonal' axes offer two possible settings called 'obverse' and 'reverse'; obverse is the conventional one. The settings differ in the kind of centring of the hexagonal cell, namely $\pm(\frac{2}{3}, \frac{1}{3}, \frac{1}{3})$ for obverse and $\pm(\frac{1}{3}, \frac{2}{3}, \frac{1}{3})$ for reverse. Sometimes cell transformations can be avoided if the reverse setting is chosen. For example, when a rhombohedral space group results from another rhombohedral space group by doubling of the (hexagonal) c axis, either the directions of **a** and **b** have to be reversed or one can keep the directions if one of the two space groups is chosen with a reverse setting. Since this cannot be expressed by the Hermann–Mauguin symbol, it has to be explicitly mentioned, preferably by a superscript ^(rev) after the corresponding Hermann–Mauguin symbol.

9.3.5 Hexagonal space groups

In former times, hexagonal space groups were described with an H cell or a C cell. These result from the conventional cell according to:

Basis vectors of the H cell: $2\mathbf{a} + \mathbf{b}$, $-\mathbf{a} + \mathbf{b}$, $\mathbf{c}$

Basis vectors of the C cell: $2\mathbf{a} + \mathbf{b}$, $\mathbf{b}$, $\mathbf{c}$ or $\mathbf{a}$, $\mathbf{a} + 2\mathbf{b}$, $\mathbf{c}$

The C cell (orthohexagonal cell) corresponds to the cell of orthorhombic subgroups of hexagonal space groups. The H cell is centred in the positions $\pm(\frac{2}{3}, \frac{1}{3}, 0)$. Both kinds of cell are usually of little advantage and are used only seldom.

Exercises

Solutions in Appendix D (page 287)

- (9.1) The cell of space group $P\bar{1}$ is to be converted to another cell with the basis vectors **a**, **b**, **a** + **b** + **c**. How do the Wyckoff symbols have to be interchanged? Set up the transformation matrices **P** and $\mathbf{P}^{-1}$. The Wyckoff symbols are: $1a(0, 0, 0)$; $1b(0, 0, \frac{1}{2})$; $1c(0, \frac{1}{2}, 0)$; $1d(\frac{1}{2}, 0, 0)$; $1e(\frac{1}{2}, \frac{1}{2}, 0)$; $1f(\frac{1}{2}, 0, \frac{1}{2})$; $1g(0, \frac{1}{2}, \frac{1}{2})$; $1h(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$; $2i(x, y, z)$.
- (9.2) What are the Hermann–Mauguin symbols of space group $P2_1/n2_1/m2_1/a$ for the axes settings **c**, **a**, **b** and **b**, **a**, **c**? What are the corresponding coordinates of the point $0.24, \frac{1}{4}, 0.61$?
- (9.3) What has to be considered when the axes **a** and **b** of the space group $C12/c1$ are interchanged?
- (9.4) What is the Hermann–Mauguin symbol of the space group $P4_2/n2_1/c2/m$ with a *C*-centred setting?
- (9.5) Among others, the following supergroups of space group $P2_1/c$ ($P12_1/c1$, No. 14) are listed in *International Tables* with their standard symbols; the full symbols have been added here in parentheses:
- $Pnna(P2/n2_1/n2/a)$; $Pcca(P2_1/c2/c2/a)$;
 $Pccn(P2_1/c2_1/c2/n)$; $Cmce(C2/m2/c2_1/e)$.

Only the standard symbols are listed for supergroups, even if this requires a basis transformation that is not mentioned. What are the Hermann–Mauguin symbols of the supergroups if their axes are oriented in the same way as for $P12_1/c1$? Is it necessary, in some of the cases, to choose some other setting of $P12_1/c1$ in order to be able to keep the orientation of the axes for the supergroup? What conditions must be met by the cell of $P12_1/c1$ so that the mentioned space groups can actually be supergroups?